Manipal Institute of Technology, MAHE

SOFTWARE ENGINEERING LAB
(CSE 3141)

PORTKEY
Food Delivery System

Project Report

Submitted By:

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1 Scope of the Project

The PORTKEY Food Delivery System aims to provide customers with the comfort of ordering food from home or quickly identifying nearby restaurants for dine-in. Users can browse local restaurants, order food, make payments, contact the restaurant or delivery person, and track deliveries in real-time.

Key Features:
• User registration
• Restaurant listing with menus
• Online food ordering system
• Order tracking
• Restaurant and Delivery partner contact information
• Payment integration (COD/Online payment services)
• Admin dashboard for order history, contact details, etc.

2 Objective

To provide easy and fast access to food anytime as per user convenience with complete restaurant information for dine-in, take-outs, or home delivery.

Key Objectives:
• Design a robust, user-friendly food delivery platform.
• Reduce manual customer support load using an ML chatbot.
• Enhance user experience through conversational automation.
• Ensure secure and efficient food ordering and tracking.
• Provide 24x7 intelligent query resolution.

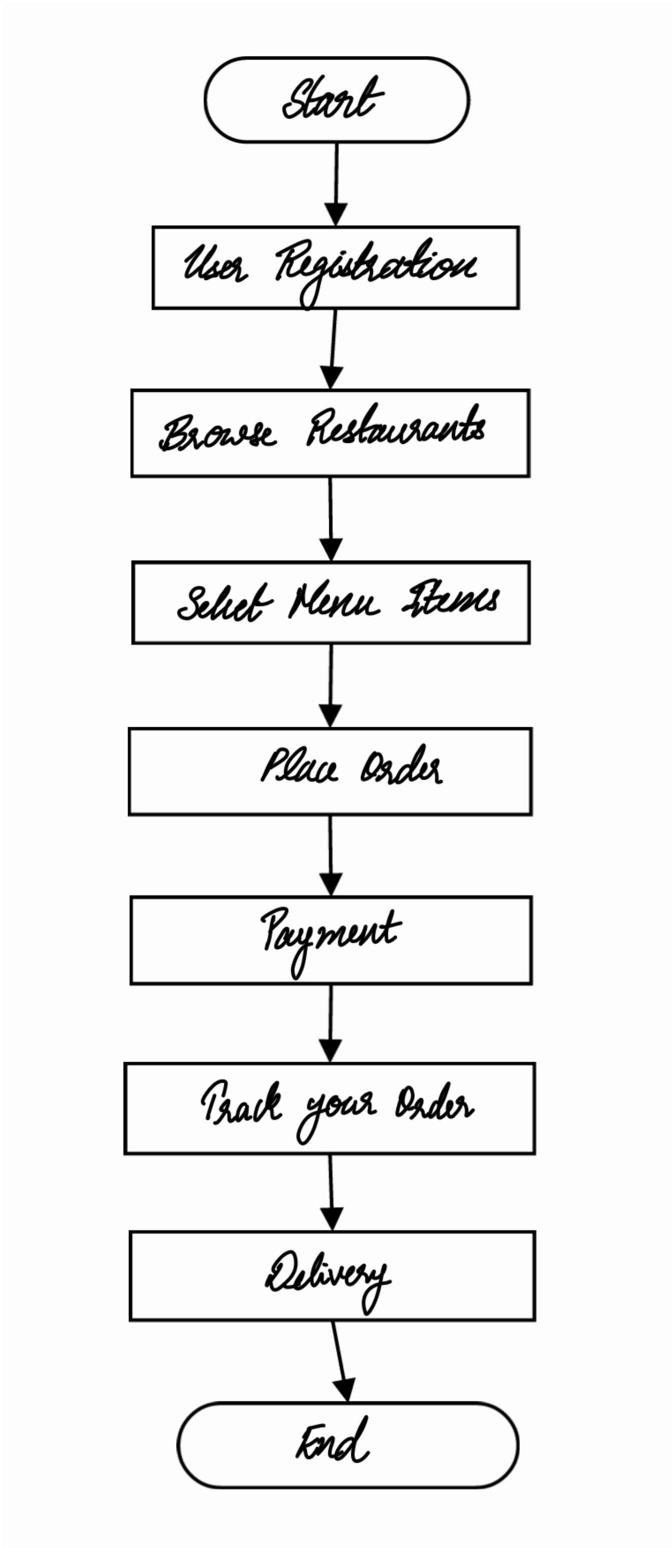
3 Hardware and Software Requirements

Hardware Requirements:
• Minimum 8 GB RAM
• Intel i3, i5 processor or above
• 500 GB hard disk
• Stable Internet Connection

Software Requirements:
Frontend: HTML, CSS, JavaScript
Backend: Node.js / Flask / Django
Database: MySQL / MongoDB
Chatbot: LLM Based
Version Control: Git, GitHub
OS: Windows

4 Flowchart and ER Diagram

Flowchart



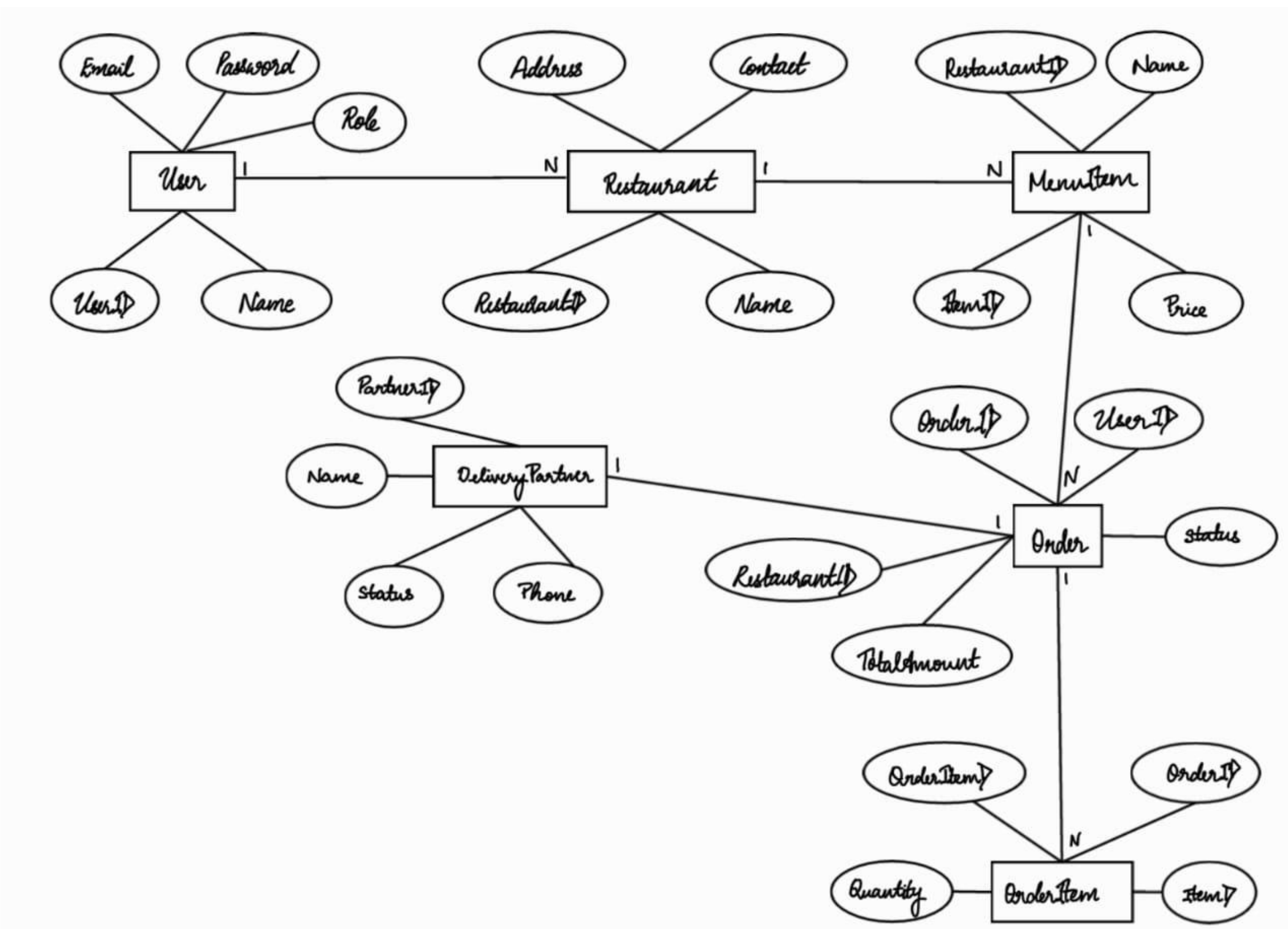
ER Diagram

Entities:

• User (UserID, Name, Email, Password, Role)
• Restaurant (RestaurantID, Name, Address, Contact)
• MenuItem (ItemID, RestaurantID, Name, Price)
• Order (OrderID, UserID, RestaurantID, TotalAmount, Status)
• OrderItem (OrderItemID, OrderID, ItemID, Quantity)
• DeliveryPartner (PartnerID, Name, Phone, Status)

Relationships:

• A User can place many Orders
• A Restaurant has many MenuItems
• An Order contains multiple OrderItems
• A DeliveryPartner is assigned to one Order



5 Tasks Involved

1. Requirement gathering – All
2. Design (UI, ERD, Flowchart) – All
3. Frontend development (UI, menu, order flow) – Abhay
4. Backend APIs (login, order, cart, delivery) – Rithik
5. Database schema & integration – Rithik
6. Payment gateway simulation – Abhay
7. Delivery module logic – Princita
8. Testing & debugging – All
9. Final Report & Presentation – All

6 Individual Contributions

Abhay: UI/UX Design, Frontend Pages, Order Flow, Tracking UI
Rithik: Backend Logic, API Development, Payment Integration
Princita: Data processing, Chatbot Integration